

ECE 414 / Spring 2026 / HW3

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Corrections shown in red. Every problem was checked against the solution key. The only error was in Problem 4(d): a missing factor of 2π when converting the Ramsey linewidth ($\sim 1/T$, an *angular* frequency) into Hz. This propagates to the fractional frequency stability.

Problem 1

Use the reduced two-level system $|1\rangle \leftrightarrow |2\rangle$:

$$\tilde{H} = \hbar \begin{pmatrix} 0 & \Omega/2 \\ \Omega/2 & -\delta \end{pmatrix}.$$

Here Γ is spontaneous decay $|2\rangle \rightarrow |1\rangle$, and γ is incoherent pumping $|1\rangle \rightarrow |2\rangle$.

(a)

$$\dot{\rho}_{11} = \Gamma\rho_{22} - \gamma\rho_{11} + \frac{i\Omega}{2}(\rho_{12} - \rho_{21}), \quad (1)$$

$$\dot{\rho}_{22} = -\Gamma\rho_{22} + \gamma\rho_{11} - \frac{i\Omega}{2}(\rho_{12} - \rho_{21}), \quad (2)$$

$$\dot{\tilde{\rho}}_{12} = \frac{i\Omega}{2}(\rho_{11} - \rho_{22}) - \left(\frac{\Gamma+\gamma}{2} + i\delta\right)\tilde{\rho}_{12}, \quad (3)$$

$$\dot{\tilde{\rho}}_{21} = -\frac{i\Omega}{2}(\rho_{11} - \rho_{22}) - \left(\frac{\Gamma+\gamma}{2} - i\delta\right)\tilde{\rho}_{21}. \quad (4)$$

(b) Define

$$W = \rho_{22} - \rho_{11}, \quad a = \frac{\Gamma + \gamma}{2}.$$

At steady state,

$$\tilde{\rho}_{12}^{ss} = -\frac{i\Omega W_{ss}}{2(a + i\delta)}, \quad \boxed{W_{ss} = \frac{\gamma - \Gamma}{(\gamma + \Gamma) + \Omega^2 a / (a^2 + \delta^2)}}.$$

Thus

$$\rho_{22}^{ss} = \frac{1 + W_{ss}}{2}, \quad \rho_{11}^{ss} = \frac{1 - W_{ss}}{2}.$$

For a weak probe,

$$\rho_{22}^{(0)} = \frac{\gamma}{\gamma + \Gamma}, \quad \rho_{11}^{(0)} = \frac{\Gamma}{\gamma + \Gamma}.$$

(c) Absorption: $\rho_{22} < \rho_{11}$. Gain: $\rho_{22} > \rho_{11}$. Therefore

$$\boxed{\gamma > \Gamma}$$

is needed for gain in the weak-probe limit.

(d)

$$G = \xi W_{ss}, \quad G = \eta \quad \text{at steady state.}$$

Therefore

$$\Omega_{ss}^2 = \frac{a^2 + \delta^2}{a} \left[\frac{\xi(\gamma - \Gamma)}{\eta} - (\gamma + \Gamma) \right].$$

Since $I_{\text{in}} = \beta \Omega_{ss}^2$,

$$I_{\text{out}} = (1 - R_{\text{out}}) \beta \Omega_{ss}^2.$$

For $\gamma = 5\Gamma$, $a = 3\Gamma$, and $R_{\text{out}} = 0.9$,

$$I_{\text{out}} = 0.1 \beta (\delta^2 + 9\Gamma^2) \left(\frac{4\xi}{3\eta} - 2 \right).$$

Threshold:

$$\xi \frac{\gamma - \Gamma}{\gamma + \Gamma} = \eta \Rightarrow \gamma_{\text{min}} = \Gamma \frac{\xi + \eta}{\xi - \eta}.$$

Problem 2

(a)

$$\tilde{H}' = \hbar \begin{pmatrix} 0 & \Omega_0/2 \\ \Omega_0/2 & -\delta - i\Gamma/2 \end{pmatrix}.$$

Since $\tilde{H}' \neq \tilde{H}'^\dagger$,

$$\dot{\rho}_{\text{Ham}} = -\frac{i}{\hbar} \left(\tilde{H}' \tilde{\rho} - \tilde{\rho} \tilde{H}'^\dagger \right).$$

(b) Add the source term $\dot{\rho}_{11}|_{\text{source}} = \Gamma \rho_{22}$. Then

$$\dot{\rho}_{11} = \Gamma \rho_{22} + \frac{i\Omega_0}{2} (\rho_{12} - \rho_{21}), \quad (5)$$

$$\dot{\rho}_{22} = -\Gamma \rho_{22} - \frac{i\Omega_0}{2} (\rho_{12} - \rho_{21}), \quad (6)$$

$$\dot{\rho}_{12} = \frac{i\Omega_0}{2} (\rho_{11} - \rho_{22}) - \left(\frac{\Gamma}{2} + i\delta \right) \tilde{\rho}_{12}, \quad (7)$$

$$\dot{\rho}_{21} = -\frac{i\Omega_0}{2} (\rho_{11} - \rho_{22}) - \left(\frac{\Gamma}{2} - i\delta \right) \tilde{\rho}_{21}. \quad (8)$$

(c)

$$\tilde{H}' = \tilde{H} + \bar{D}, \quad \bar{D} = -\frac{i\hbar}{2} \begin{pmatrix} 0 & 0 \\ 0 & \Gamma \end{pmatrix}.$$

The decay part is

$$\dot{\rho}_{\text{decay}} = -\frac{i}{\hbar} \left(\bar{D} \rho - \rho \bar{D}^\dagger \right).$$

(d) For the three-level system,

$$\tilde{H} = \hbar \begin{pmatrix} \Delta/2 & 0 & \Omega_a/2 \\ 0 & -\Delta/2 & \Omega_b/2 \\ \Omega_a/2 & \Omega_b/2 & -\delta \end{pmatrix}, \quad \Gamma = \gamma_a + \gamma_b.$$

Use

$$\bar{D} = -\frac{i\hbar}{2} \begin{pmatrix} \gamma_d & 0 & 0 \\ 0 & \gamma_c & 0 \\ 0 & 0 & \Gamma \end{pmatrix}.$$

Population source:

$$\dot{\rho}_{\text{source}} = \begin{pmatrix} \gamma_a \rho_{33} + \gamma_c \rho_{22} & 0 & 0 \\ 0 & \gamma_b \rho_{33} + \gamma_d \rho_{11} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

So

$$\dot{\rho} = -\frac{i}{\hbar} [\tilde{H}, \rho] - \frac{i}{\hbar} (\bar{D}\rho - \rho\bar{D}^\dagger) + \dot{\rho}_{\text{source}}.$$

The main components are

$$\dot{\rho}_{11} = -\frac{i\Omega_a}{2}(\rho_{31} - \rho_{13}) - \gamma_d \rho_{11} + \gamma_c \rho_{22} + \gamma_a \rho_{33}, \quad (9)$$

$$\dot{\rho}_{22} = -\frac{i\Omega_b}{2}(\rho_{32} - \rho_{23}) - \gamma_c \rho_{22} + \gamma_d \rho_{11} + \gamma_b \rho_{33}, \quad (10)$$

$$\dot{\rho}_{33} = -\frac{i\Omega_a}{2}(\rho_{13} - \rho_{31}) - \frac{i\Omega_b}{2}(\rho_{23} - \rho_{32}) - \Gamma \rho_{33}, \quad (11)$$

$$\dot{\rho}_{12} = -i \left[\Delta \rho_{12} + \frac{\Omega_a}{2} \rho_{32} - \frac{\Omega_b}{2} \rho_{13} \right] - \frac{\gamma_c + \gamma_d}{2} \rho_{12}, \quad (12)$$

$$\dot{\rho}_{13} = -i \left[\left(\delta + \frac{\Delta}{2} \right) \rho_{13} + \frac{\Omega_a}{2} (\rho_{33} - \rho_{11}) - \frac{\Omega_b}{2} \rho_{12} \right] - \frac{\gamma_d + \Gamma}{2} \rho_{13}, \quad (13)$$

$$\dot{\rho}_{23} = -i \left[\left(\delta - \frac{\Delta}{2} \right) \rho_{23} + \frac{\Omega_b}{2} (\rho_{33} - \rho_{22}) - \frac{\Omega_a}{2} \rho_{21} \right] - \frac{\gamma_c + \Gamma}{2} \rho_{23}. \quad (14)$$

(e) If $\gamma_c = \gamma_d = 0$, $\gamma_a = \gamma_b = \Gamma/2$,

$$\dot{\rho}_{11}|_{\text{source}} = \dot{\rho}_{22}|_{\text{source}} = \frac{\Gamma}{2} \rho_{33}.$$

This is the symmetric result.

(f) If $\Omega_a = \Omega_b = 0$,

$$\boxed{\rho_{11}^{ss} = \frac{\gamma_c}{\gamma_c + \gamma_d}, \quad \rho_{22}^{ss} = \frac{\gamma_d}{\gamma_c + \gamma_d}, \quad \rho_{33}^{ss} = 0.}$$

(g) Thermal equilibrium gives

$$\frac{\rho_{22}}{\rho_{11}} = e^{-\hbar(\omega_2 - \omega_1)/(k_B T)} = \frac{\gamma_d}{\gamma_c}.$$

Thus

$$\boxed{\frac{\gamma_c}{\gamma_d} = e^{\hbar(\omega_2 - \omega_1)/(k_B T)}.$$

For $T = 295$ K, $\omega_2 - \omega_1 = 2\pi(6.8 \times 10^9) \text{ s}^{-1}$,

$$\boxed{\gamma_c/\gamma_d \approx 1.0011.}$$

Problem 3

$$\dot{\tilde{C}}_1 = -i\frac{\Omega_0}{2}\tilde{C}_2, \quad \dot{\tilde{C}}_2 = -i\frac{\Omega_0}{2}\tilde{C}_1 + \left(i\delta - \frac{\Gamma}{2}\right)\tilde{C}_2.$$

(a) Set $\dot{\tilde{C}}_2 \simeq 0$:

$$\tilde{C}_2 \simeq \frac{\Omega_0}{2\delta + i\Gamma}\tilde{C}_1.$$

If $|\tilde{C}_1| \simeq 1$,

$$\boxed{\rho_{22} \simeq \frac{\Omega_0^2}{4\delta^2 + \Gamma^2} \simeq \frac{\Omega_0^2}{4\delta^2}.$$

(b) For $\delta \gg \Gamma$,

$$\tilde{C}_2 \simeq \frac{\Omega_0}{2\delta}\tilde{C}_1, \quad \dot{\tilde{C}}_1 = -i\frac{\Omega_0^2}{4\delta}\tilde{C}_1.$$

With $\tilde{C}_1(0) = 1$,

$$\boxed{\tilde{C}_1(t) = e^{-i\Omega_0^2 t/(4\delta)}, \quad \tilde{C}_2(t) \simeq \frac{\Omega_0}{2\delta}e^{-i\Omega_0^2 t/(4\delta)}.$$

(c)

$$R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad \tilde{H} = \frac{\hbar}{2} \begin{pmatrix} 0 & \Omega_0 \\ \Omega_0 & -2\delta \end{pmatrix}.$$

Diagonalization requires

$$\boxed{\tan(2\theta) = \frac{\Omega_0}{\delta}.$$

Let $\Omega' = \sqrt{\Omega_0^2 + \delta^2}$. Then

$$\tilde{H}' = \hbar \begin{pmatrix} \alpha_A & 0 \\ 0 & \alpha_B \end{pmatrix}, \quad \alpha_A = \frac{\Omega' - \delta}{2}, \quad \alpha_B = \frac{-\Omega' - \delta}{2}.$$

For $\delta \gg \Omega_0$,

$$\boxed{\theta \simeq \frac{\Omega_0}{2\delta}, \quad \alpha_A \simeq \frac{\Omega_0^2}{4\delta}, \quad \alpha_B \simeq -\delta - \frac{\Omega_0^2}{4\delta}.$$

Thus the same result as part (b) is obtained.

Problem 4

$$i\hbar \frac{d}{dt} \begin{pmatrix} \tilde{C}_1 \\ \tilde{C}_2 \\ \tilde{C}_3 \end{pmatrix} = \hbar \begin{pmatrix} \Delta/2 & 0 & \Omega_a/2 \\ 0 & -\Delta/2 & \Omega_b/2 \\ \Omega_a/2 & \Omega_b/2 & -\delta - i\Gamma/2 \end{pmatrix} \begin{pmatrix} \tilde{C}_1 \\ \tilde{C}_2 \\ \tilde{C}_3 \end{pmatrix}.$$

(a) For $\Omega_a = \Omega_b = \Omega_0$, set $\dot{\tilde{C}}_3 \simeq 0$:

$$\boxed{\tilde{C}_3 \simeq \frac{\Omega_0}{2(\delta + i\Gamma/2)}(\tilde{C}_1 + \tilde{C}_2) \simeq \frac{\Omega_0}{2\delta}(\tilde{C}_1 + \tilde{C}_2).$$

(b) After eliminating $|3\rangle$,

$$\tilde{H}_{\text{eff}} = \hbar \begin{pmatrix} \Delta/2 + \Omega_0^2/(4\delta) & \Omega_0^2/(4\delta) \\ \Omega_0^2/(4\delta) & -\Delta/2 + \Omega_0^2/(4\delta) \end{pmatrix}.$$

Drop the common light shift and define $\beta = \frac{\Omega_0^2}{2\delta}$. Then

$$\tilde{H}' = \hbar \begin{pmatrix} \Delta/2 & \beta/2 \\ \beta/2 & -\Delta/2 \end{pmatrix}.$$

(c) With

$$Q = \begin{pmatrix} e^{-i\theta_1 t} & 0 \\ 0 & e^{i\theta_2 t} \end{pmatrix},$$

choose $\theta_1 + \theta_2 = 0$.

$$\theta_1 = -\beta/2, \quad \theta_2 = \beta/2$$

gives

$$\tilde{H}' = \hbar \begin{pmatrix} \Delta/2 & \beta/2 \\ \beta/2 & -\Delta/2 \end{pmatrix}.$$

Also,

$$\theta_1 = -(\Delta + \beta)/2, \quad \theta_2 = (\Delta + \beta)/2$$

gives

$$\tilde{H}'' = \hbar \begin{pmatrix} 0 & \beta/2 \\ \beta/2 & -\Delta \end{pmatrix}.$$

(d) Use

$$\rho'_{12} = \frac{1}{2}(R_1 + iR_2), \quad \rho'_{11} - \rho'_{22} = R_3.$$

The effective field is $(\beta, 0, \Delta)$. For two $\pi/2$ pulses,

$$\tau = \frac{\pi}{2\beta},$$

and dark time T ,

$$\rho'_{22}(\Delta) = \frac{1}{2}[1 + \cos(\Delta T)].$$

The linewidth is of order $1/T$. With $T = 10^{-3}$ s and $\text{SNR} = 10^4$:

$$\Delta f \simeq \frac{1/T}{\text{SNR}} = 0.10 \text{ Hz.} \quad \Delta f = \frac{1/T}{2\pi \text{SNR}} = \frac{1}{2\pi T \text{SNR}} = \frac{1}{2\pi(10^{-3})(10^4)} = 0.016 \text{ Hz.}$$

The linewidth $1/T$ is an *angular* frequency (rad/s); dividing by 2π converts it to Hz. The factor of 2π was missing.

For $f = 6.8 \times 10^9$ Hz:

$$\frac{\Delta f}{f} \simeq 1.47 \times 10^{-11}. \quad \frac{\Delta f}{f} = \frac{0.016}{6.8 \times 10^9} \simeq 2.35 \times 10^{-12}.$$

Problem 5

(a) Resonance condition:

$$\hbar\omega = \hbar\omega_{21} + \frac{\hbar^2 k^2}{2M}, \quad k = \frac{\omega}{c}.$$

Thus

$$\omega = \omega_{21} + \frac{\hbar\omega^2}{2Mc^2}, \quad \omega_{21} = 2\pi(5 \times 10^{14}) \text{ s}^{-1}.$$

For ^{87}Rb , $M = 87 \text{ amu} = 1.4447 \times 10^{-25} \text{ kg}$:

$$\omega \approx 3.14159265363 \times 10^{15} \text{ s}^{-1}, \quad f \approx 5.000000000064 \times 10^{14} \text{ Hz},$$

$$k \approx 1.048 \times 10^7 \text{ m}^{-1}.$$

(b)

$$v_r = \frac{\hbar k}{M} \approx 7.65 \times 10^{-3} \text{ m/s}.$$

The area is

$$A = v_x v_r T^2.$$

With $v_x = 300 \text{ m/s}$, $T = 3 \text{ ms}$,

$$A \approx 2.07 \times 10^{-5} \text{ m}^2.$$

(c) The total interferometer phase is

$$\Phi = (\phi_a + \phi_c) - (\phi_b + \phi_d).$$

After the last $\pi/2$ pulse,

$$\rho_{11}^{\text{out}} = \frac{1}{2}[1 + \cos \Phi] = \frac{1}{2}[1 + \cos(\phi_a + \phi_c - \phi_b - \phi_d)].$$

(d) For the Raman version,

$$|\tilde{C}_3|^2 \simeq \frac{\Omega_0^2}{4\delta^2} = \epsilon.$$

Also,

$$\beta\tau = \frac{\pi}{2}, \quad \beta = \frac{\Omega_0^2}{2\delta}.$$

So

$$\delta = \frac{\pi}{4\epsilon\tau}, \quad \Omega_0 = \frac{\pi}{2\tau\sqrt{\epsilon}}.$$

For $\tau = 3 \times 10^{-6} \text{ s}$, $\epsilon = 0.01$,

$$\delta = 2.62 \times 10^7 \text{ s}^{-1} = 2\pi(4.17 \text{ MHz}),$$

$$\Omega_0 = 5.24 \times 10^6 \text{ s}^{-1} = 2\pi(0.833 \text{ MHz}).$$

(e) In a Raman transition,

$$k_{\text{eff}} = k_a + k_b \simeq 2k.$$

Therefore,

$$A_{\text{Raman}} \simeq 2A \approx 4.13 \times 10^{-5} \text{ m}^2.$$